Fifth Industrial Revolution

Human progress has always been shaped by technology. Electricity transformed how we lived and worked. Computers changed how we processed information, and the internet connected the world. Later, mobile phones brought technology into the hands of everyone.

Each phase reshaped society. We are now entering the Fifth Industrial Revolution, where AI works alongside humans to transform every sector of the economy. It builds on the digital advances of the past decade. It focuses on humans and machines working together, using AI to improve lives and address challenges in agriculture, health, climate change and a host of other sectors.

AI a Multiplier

AI is often described as a technology enabler with a multiplier effect because it strengthens existing systems rather than operating in isolation. When used in areas like farming, healthcare, manufacturing or governance, it makes processes faster, more efficient and accurate.

Improvement in one sector creates multiple benefits across the economy, including higher productivity, lower costs and better decisions. Countries that develop and use AI effectively stand to gain an advantage in competitiveness and resilience. In this sense, AI is becoming a capability that strengthens the nation and improves the lives of its people.

Our prime minister has consistently emphasised that technology in India must be democratised, both in usage and in development. It should not be confined to a few companies, individuals or even nations. Every section of society must benefit from it.

India's digital revolution also reflects this philosophy. UPI transformed digital payments by making them open and accessible. Aadhaar enabled digital identity at scale. India built its own 4G/5G stack. AI must also follow the same path.



TAKEAWAYS

- **India is among the top three startup ecosystems. Today, nearly 90% of startups are AI-powered in some form**
- **India needs sovereign models to ensure data security, strategic autonomy; IndiaAI Mission has 12 AI models going**
- **In India, AI is used across sectors like agriculture, healthcare, education, manufacturing, governance, climate action**

AI Stack: Application First

In our understanding, the AI stack has five layers. At the top of this stack is the application layer. In the 1960s and '70s, semiconductors laid the foundation of modern electronics. But the actual impact was realised when these were widely adopted across industries. Similarly, the internet created value when people and systems began using it at scale. Mobile phones transformed society when they became part of everyday life. Sensors delivered real impact once they were widely deployed to observe and measure the physical world. The data they generated was used to solve real-world problems.

AI will follow the same pattern.